

Trustworthy and Resilient Network Intrusion Detection

A reproducible pilot under false-alarm and computational constraints

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PURPOSE

This compact study demonstrates the ability to move from a doctoral research idea to an auditable experiment. It is a proof-of-execution artifact, not a completed paper and not a claim of a novel classifier.

Motivation

High detection scores alone do not make an intrusion detector deployable. False alarms, missed attacks, inference cost, benchmark leakage, and uncertainty all affect operational usefulness. This pilot therefore treats model selection as a constrained decision problem rather than an accuracy contest.

Research question

Can validation-only operating-point selection produce a more useful trade-off among attack recall, false-positive rate, model size, and inference cost than default-threshold baselines?

Core result

A leakage-controlled audit and baseline comparison selected a Random Forest threshold of **0.59853** using validation only. On held-out test data, attack recall was **98.167%** and FPR was **1.208%**. The 95% FPR interval was **0.846% to 1.601%**; therefore, the experiment does not establish that FPR is reliably below the 1% operational target.

The key research insight is methodological: satisfying a point constraint on validation did not guarantee constraint satisfaction on held-out data. This motivates confidence-bounded adaptive selection under distribution shift.

Dataset and provenance

CICIoT2023 was retrieved from the official Canadian Institute for Cybersecurity portal. The producer describes 105 IoT devices and 33 attacks in seven categories. The pilot is pinned to **Merged01.csv** (712,311 rows; 39 predictors) with SHA-256 8b43d6552a8cafd3b0ca2cedf6464ca3fe644d7fc9bb5dfe906368f1542792fe. Raw data are not redistributed.

G1 - Audit before modeling

Finding	Observed	Methodological consequence
Exact duplicate rows	167,368 (23.50%)	Ordinary random splitting risks leakage
Duplicate feature rows	194,204 (27.26%)	Feature groups must remain within one partition
Conflicting binary-label groups	16	Excluded from the primary analysis
Attack prevalence	97.673%	Accuracy and attack PR-AUC are insufficient
Missing / infinite cells	22 / 14	Training-only imputation after infinity conversion

Methodology

After label normalization, conflicting feature groups were removed and one representative per exact feature vector was retained. The remaining 518,091 observations were split 60/20/20 into development, validation, and test partitions with seed 20260920. Preprocessing was fitted on development data only.

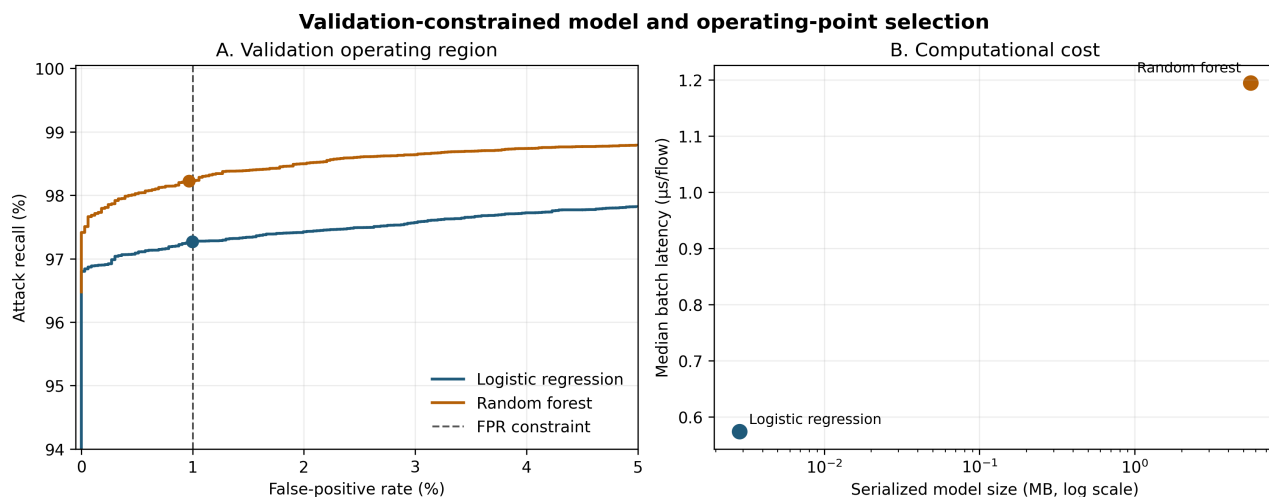
Two class-weighted baselines were implemented: Logistic Regression and Random Forest. The contribution is a validation-only decision rule: maximize attack recall subject to FPR $\leq 1\%$; if recall differs by no more than 0.1 percentage points, select the smaller serialized model. The chosen rule and threshold were frozen before test application.

G2 - Baselines at threshold 0.50

Test metric	Logistic Regression	Random Forest
Attack recall	96.880%	98.391%
FPR	0.030%	1.843%
MCC	0.705	0.803
Macro average precision	0.885	0.946
Model size	2.9 KB	5.56 MB
Median batch latency	0.574 us/flow	1.195 us/flow

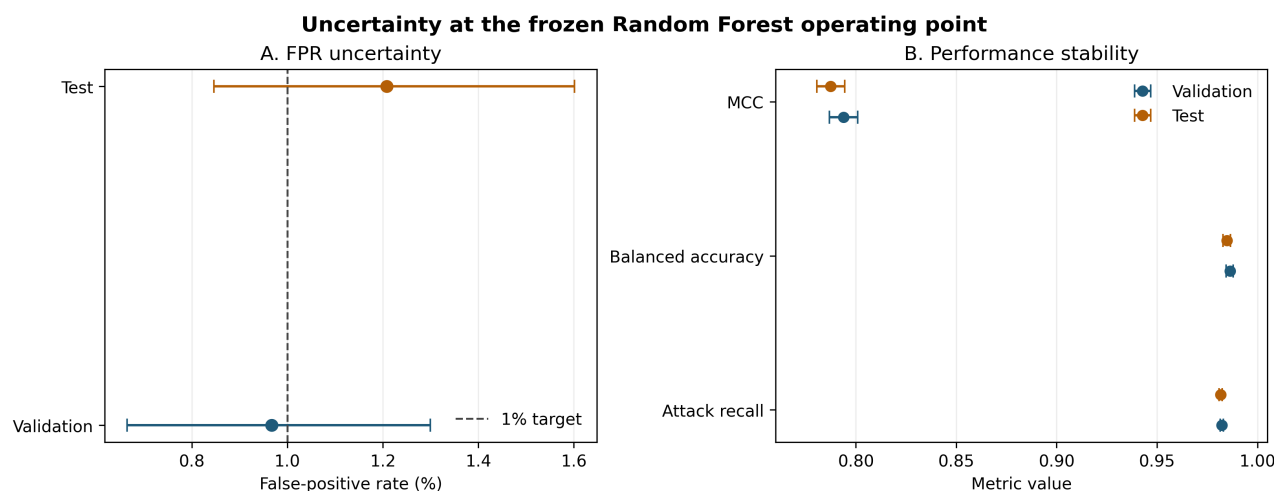
No model dominates every objective: Random Forest detects more attacks, while Logistic Regression produces fewer false alarms and is substantially smaller.

G3 - Validation-constrained selection



Random Forest was selected at threshold 0.59853 because validation recall (98.228%) exceeded the Logistic Regression feasible point (97.266%) by more than the pre-specified near-tie tolerance. On test, the frozen threshold reduced false positives from 61 to 40 relative to the default Random Forest threshold, but increased missed attacks from 1,614 to 1,839.

G4 - Uncertainty at the frozen point



Held-out metric	Observed	95% bootstrap interval
False-positive rate	1.2085%	0.8459% to 1.6012%
Attack recall	98.1667%	98.0839% to 98.2484%
Balanced accuracy	98.4791%	98.2851% to 98.6620%
MCC	0.7875	0.7806 to 0.7945

Interpretation and limitations

The results support reproducible performance within a leakage-controlled split of one fixed benchmark shard. They do not establish a guaranteed FPR below 1%, temporal robustness, production readiness, or cross-network generalization. The original merged dataset was globally shuffled, so this pilot cannot simulate a genuine future deployment period.

Proposed doctoral extension

Develop confidence-bounded and preference-aware operating-point adaptation for resilient intrusion detection under device, traffic, and resource shifts. Evaluate it with temporal or device-group separation, an external dataset, runtime monitoring, and explicit false-alarm and compute budgets.

Supervisor alignment pathways

Research pathway	Pilot connection
Wireless / IoT resilience	Resource-aware inference, false alarms, edge-oriented extension
Multi-objective optimization	Pareto trade-offs among recall, FPR, size, and latency
Verification / dependability	Testable constraints and failure of validation-only assurance
Trustworthy adaptive security	Leakage control, frozen evaluation, uncertainty-aware adaptation
Adaptive ML / AutoML	Confidence-bounded thresholds and model selection under shift

Reproducibility statement

The package contains a staged Python notebook, fixed random seed, pinned requirements, locked data hash, explicit split and preprocessing rules, serialized baselines, machine-readable results, and publication-quality figures. Every major statement is separated into implemented, observed, inferred, or proposed work.

References

[1] Neto, E. C. P. et al. CICIoT2023: A Real-Time Dataset and Benchmark for Large-Scale Attacks in IoT Environment. Sensors 23(13), 5941 (2023). <https://doi.org/10.3390/s23135941>

[2] Canadian Institute for Cybersecurity, University of New Brunswick. CIC IoT Dataset 2023. <https://www.unb.ca/cic/datasets/iotdataset-2023.html>

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